

220432/180435/120435

/030435/094741

**Subject : Fuel & Furnaces/ Fuel Refractory
& Furnaces**

Time : 3 Hrs.

M.M. : 60

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which of the following is solid fuel?

- a) Coal b) LPG
c) CNG d) Kerosene

Q.2 Combustion is an _____ reaction.

- a) Exothermic b) Endothermic
c) Chain d) Nuclear

Q.3 A bomb calorimeter measures _____ of fuels.

- a) Dew point b) Calorific value
c) Density d) Strength

Q.4 Which of the following is part of kiln.

- a) Chimney b) Fire box
c) Damper d) All of these

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Q.5 The primary component of air by volume is:

- a) Oxygen b) Nitrogen
- c) Argon d) CO₂

Q.6 Which of these is non Combustible?

- a) Peat b) Coke
- c) Coal d) Stone

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Coke is a by-product of coal (True/False)

Q.8 A bomb calorimeter measures the calorific value.(True/False)

Q.9 Combustion requires oxygen (True/False)

Q.10 A chimney generates natural draft (True/False)

Q.11 Tunnel kilns operates in a batch mode(True/False)

Q.12 Orsat analysis measures fuel composition (True/False)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Compare solid and gaseous fuels.

Q.14 Explain the flash point and fire point.

Q.15 Explain the role of damper in a furnace?

Q.16 Differentiate between recuperative and regenerative furnaces.

Q.17 Explain the term calorific value.

Q.18 What is pulverized coal?

Q.19 List the advantages of liquid fuels.

Q.20 Define and explain octane and cetane numbers.

Q.21 Explain the working of a thermocouple in a furnace?

Q.22 Explain the use of an Orsat apparatus.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Describe the downdraft kiln process with a diagram.

Q.24 Explain proximate analysis.

Q.25 Describe the calorific value determination in a bomb calorimeter.

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